SpecialtyCare · Greater Boston neuromonitoring team

IONM Team Publications

15 papers and abstracts on intraoperative neuromonitoring, with plain-language summaries. 7 are free to read. Titles and links in this PDF are clickable.

Updated October 2026.

With Mass General Brigham Neurosurgery 12

Cases, techniques and reviews co-authored with neurosurgeons at Brigham and Women's Hospital and Mass General Brigham.

2026 · CASE REPORT · FREE FULL TEXT

Delayed identification and treatment of a cervicomedullary spinal vascular lesion following angiogram-negative, high-grade subarachnoid hemorrhage: illustrative case

Chang C, Motoc M, Can A, **Lachs M**, **Bose M**, Bernstock JD, Patel NJ

J Neurosurg Case Lessons. 2026;11(26):CASE26229. doi:10.3171/case26229 · PMID 42372299

A 54-year-old man with high-grade subarachnoid hemorrhage and an initially negative angiogram was later found to have an aneurysm and AVM beside the anterior spinal artery at the cervicomedullary junction. Endovascular treatment was judged too risky, so the lesion was resected through a far-lateral transcondylar approach. A decline in MEPs prompted the team to reinspect and correct an inadvertent clip-related injury, and the lesion was obliterated without new deficits. The authors recommend repeat imaging when initial studies are negative but cervicomedullary clues are present.

MONITORING TAKEAWAY An MEP decline caught a correctable clip-related problem before closure.

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2026 · SYSTEMATIC REVIEW · FREE FULL TEXT

Corticobulbar Motor Evoked Potentials: A Systematic Review of Technical Variations

Jarvis CA, **Toczylowski M**, **Bose M**, Patel RV, Wang J, McMahon JT, Bi WL

Neurosurg Pract. 2026;7(3):e000237. doi:10.1227/neuprac.0000000000000237 · PMID 42100545

A PRISMA 2020 systematic review of 72 corticobulbar MEP studies published between 2005 and 2023. Its main finding is how unsettled the technique still is: electrode montages, stimulation set-ups and the way accuracy is measured differ widely from study to study, and only a minority report sensitivity or specificity at all. Most studies used coMEPs alongside other modalities, yet few compared them directly. The authors argue that common reporting standards are what would let coMEPs spread into routine practice.

MONITORING TAKEAWAY A catalog of the coMEP montages, paradigms and warning criteria in use; useful when writing or defending a standard protocol.

Free full text (PMC) · Publisher · PubMed

2026 · MEETING ABSTRACT

Intraoperative Trigeminal Somatosensory Evoked Potential Monitoring during Skull Base Surgery

McMahon JT, **Toczylowski M**, Jarvis CA, **Bose M**, Bi WL

J Neurol Surg B Skull Base. 2026;87(Suppl 1). doi:10.1055/s-0046-1818935

Retrospective review of trigeminal SSEP monitoring in 67 skull base craniotomies (2022 to 2025). Reliable V1 to V3 responses were obtained from 96% of 159 branches, including nearly every branch in patients who already had facial numbness. Six patients had intraoperative amplitude drops of roughly 30 to 60%, half of which recovered by closure; of the 61 without changes, one developed new numbness that persisted. Responses present at the end of surgery strongly predicted no lasting deficit, and the authors call for branch-specific alert thresholds.

MONITORING TAKEAWAY Trigeminal SSEPs are obtainable in almost every branch, even with baseline numbness; an unrecovered drop may predict lasting facial deficit.

Publisher

2025 · LETTER

In Reply: Interpretation and Strategy to Resolve Neuromonitoring Changes Associated With Brain Sag

Bi WL, **Toczylowski M**, Guo X, **Bose M**

Oper Neurosurg. 2025;28(6):911. doi:10.1227/ons.0000000000001566 · PMID 40168524

The authors' response to a letter from Sampath and colleagues about their brain-sag paper. They restate the working rule from that study: signal losses caused by sag recovered reliably with saline irrigation, so a change that persists despite irrigation is treated as real. They describe their multimodal set-up (MEPs, cortical and subcortical mapping, intraoperative ultrasound), concede that sag is judged by eye and its frequency is hard to measure retrospectively, and credit refinements in electrode placement to close surgeon and neurophysiologist collaboration.

MONITORING TAKEAWAY Irrigation-responsive loss points to sag artifact; loss that persists after irrigation is treated as a true change.

Publisher · PubMed

2025 · OPERATIVE VIDEO · FREE FULL TEXT

Interhemispheric Transcallosal Approach for Resection of a Pineal Region Third Ventricular to Brainstem Tumor: 2-Dimensional Operative Video

Guo X, **Bose M**, **Toczylowski M**, Fonseca A, Bi WL

Oper Neurosurg. 2025;29(6):907-908. doi:10.1227/ons.0000000000001517 · PMID 39982058

Operative video of a 19-year-old man with a pineal-region tumor reaching from the third ventricle toward the brainstem, removed through an anterior interhemispheric transcallosal, transchoroidal corridor. Subcortical mapping of the corticospinal tracts and brainstem nuclei, together with intraoperative MRI, guided the resection, and residual nodules were taken in a planned second stage to reach gross total resection. Pathology showed a mature teratoma. He improved steadily afterward and was walking on his own by eight months, with no new visual or endocrine deficits.

MONITORING TAKEAWAY Subcortical CST and brainstem-nucleus mapping combined with iMRI in a deep midline approach.

Free full text (PMC) · Publisher · PubMed

2025 · MEETING ABSTRACT

Corticobulbar Motor Evoked Potentials for Facial and Lower Cranial Nerves: A Systematic Review of Applications and Limitations

Jarvis CA, Patel R, McMahon JT, **Toczylowski M**, **Bose M**, Bi WL

J Neurol Surg B Skull Base. 2025;86(Suppl 1). doi:10.1055/s-0045-1803714

Skull base meeting abstract that preceded the 2026 Neurosurgery Practice review. It covers 72 studies and 2,711 cases (2005 to 2023), 93% of them retrospective case reports or series with a median of 26 patients. The facial nerve was the usual target (41 studies), followed by the vagus (10) and hypoglossal (4), and about half the operations were in the cerebellopontine angle. Techniques varied widely, and reported sensitivity ranged from 40 to 100% and specificity from 60 to 100%.

MONITORING TAKEAWAY Facial nerve dominates the coMEP literature; reported accuracy is wide-ranging and methods unstandardized.

Publisher

2025 · OPERATIVE VIDEO · FREE FULL TEXT

Brain mapping during resection of high-grade brain arteriovenous malformation

Kappel AD, **Bose M**, **Toczylowski M**, Patel NJ

Neurosurg Focus Video. 2025;12(1):V7. doi:10.3171/2024.10.focvid24107 · PMID 39845307

Technique video on removing high-grade AVMs that sit in eloquent territory, here involving somatosensory cortex and the visual and corticospinal pathways. Through a bilateral craniotomy, the team stimulated cortex and subcortical white matter with a suction stimulator so the motor tracts could be followed continuously as the nidus came out, keeping roughly 3 mm from the corticospinal tract. The authors present real-time stimulation mapping as a way to make these vascular resections safer.

MONITORING TAKEAWAY Dynamic suction-stimulator mapping carried over from tumor work into AVM resection, with a ~3 mm CST margin.

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2025 · OPERATIVE VIDEO · FREE FULL TEXT

Motor mapping-guided resection of a brainstem recurrent pilocytic astrocytoma

Guo X, **Bose M**, Galvin C, Bi WL

Neurosurg Focus Video. 2025;12(1):V2. doi:10.3171/2024.10.focvid24110 · PMID 39845304

Video of a recurrent brainstem pilocytic astrocytoma removed under dynamic subcortical motor mapping with a monopolar stimulating suction device. Corticospinal responses were recorded continuously at midbrain and pontine levels, with contralateral limb and ipsilateral facial muscles captured together while tumor was removed. Resection stopped when the stimulation threshold fell below 1 mA, taken as about 1 mm or less from the tract.

MONITORING TAKEAWAY Monopolar suction-stimulator mapping with a <1 mA stop threshold and simultaneous limb and facial responses.

Free full text · **PubMed**

2025 · OPERATIVE VIDEO · FREE FULL TEXT

Multimodal-Assisted Resection of Right Insular Glioblastoma: A 2-Dimensional Operative Video

Guo X, Torio E, **Bose M**, Golby AJ, Bi WL

Oper Neurosurg. 2025;29(3):451. doi:10.1227/ons.0000000000001471 · PMID 39625292

Operative video demonstrating resection of a right insular glioblastoma with a multimodal set of intraoperative adjuncts. The full video article is free to view at the publisher.

Free full text · PubMed

2025 · ORIGINAL RESEARCH · FREE FULL TEXT

Interpretation and Strategy to Resolve Neuromonitoring Changes Associated With Brain Sag

Bose M, **Toczylowski M**, Guo X, Liu DD, Epplin-Zapf T, Wilent WB, Bi WL

Oper Neurosurg. 2025;28(6):841-854. doi:10.1227/ons.0000000000001430 · PMID 40372119

A case-based analysis from one senior surgeon's cranial tumor practice showing how brain sag, the brain shifting and settling during surgery, can produce false-positive losses of somatosensory and motor signals. The authors describe a recognizable pattern of stepwise signal decline and show that recovery during saline infusion indicates artifact rather than injury. They set out an interpretation and response strategy so the team can avoid changing the operation for a change that is not real.

MONITORING TAKEAWAY Pattern recognition for sag-related false-positive SSEP/MEP loss in cranial cases, with a stepwise response algorithm.

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2022 · OPERATIVE VIDEO

Inferior is Superior: Transtentorial Transcollateral Sulcus Approach to the Ventricular Atrium: 2-Dimensional Operative Video

Emerson SN, Kadri PADS, **Toczylowski M**, Al-Mefty O

Oper Neurosurg. 2022;23(6):e369-e370. doi:10.1227/ons.0000000000000433 · PMID 36227201

Operative video of a supracerebellar transtentorial route through the collateral sulcus to reach the ventricular atrium. The patient was a 39-year-old woman with a hemorrhagic left atrial tumor causing speech and visual symptoms; fMRI placed language in the superior temporal gyrus next to the lesion, so the inferior route was chosen to spare it. The collateral sulcus is presented as a consistent, shallow landmark that gives a short path to the atrium, with navigation used to plan the trajectory. Sacrificing a bridging vein to the tentorial sinus caused ipsilateral cerebellar edema that responded to hypertonic saline, and her vision did not worsen.

Publisher · PubMed

2022 · OPERATIVE VIDEO

Dejerine Syndrome Variant Due to Medullary Perforating Artery Ischemia During Foramen Magnum Meningioma Resection: 2-Dimensional Operative Video

Emerson SN, **Toczylowski M**, Al-Mefty O

Oper Neurosurg. 2022;23(1):e52-e53. doi:10.1227/ons.0000000000000211 · PMID 35726936

Operative video of a ventral foramen magnum meningioma resection complicated by a partial Dejerine (medial medullary) syndrome from ischemia in small perforators of the distal vertebral and anterior spinal arteries. MEPs dropped abruptly while SSEPs stayed at baseline, and postoperative MRI showed injury confined to the pyramids. The patient woke quadriplegic, needed vertebral artery stenting, and after a long recovery regained full function. The teaching point is detailed knowledge of the medullary perforators when working at the foramen magnum.

MONITORING TAKEAWAY A clean dissociation: MEPs lost, SSEPs preserved, matching an isolated pyramidal injury on MRI.

Publisher · PubMed

SpecialtyCare research 2

Studies from SpecialtyCare's clinical research group.

2020 · ORIGINAL RESEARCH

Therapeutic Impact of Traction Release After C5 Nerve Root Motor Evoked Potential (MEP) Alerts in Cervical Spine Surgery

Wilent WB, Rhee JM, Harrop JS, Epplin-Zapf T, **Bose M**, Tesdahl EA, Cohen J, Sestokas AK

Clin Spine Surg. 2020;33(10):E442-E447. doi:10.1097/bsd.0000000000000969 · PMID 32205523

A SpecialtyCare database study of 40,919 MEP-monitored cervical spine operations asking whether the surgeon's response to a C5 MEP alert changes the rate of C5 palsy. The odds of palsy rose sharply when C5 alerts occurred and persisted, and releasing traction was associated with a lower palsy risk. How sensitive the alerts were depended on the anesthetic regimen. The authors conclude that prompt traction release is an effective response, with other maneuvers chosen by context.

MONITORING TAKEAWAY Traction release as the first response to a C5 MEP alert; alert sensitivity varies with anesthetic regimen.

Publisher · PubMed

2019 · MEETING ABSTRACT

Neuroanatomical categorization of motor evoked potential alerts during cervical spine surgery and associated risk for new neurologic deficit: retrospective analysis of 40,919 procedures (Abstract 268)

Wilent WB, Epplin-Zapf TJ, Tesdahl EA, **Bose M**, Cohen J, Sestokas AK

Spine J. 2019;19(9 Suppl):S130-S131. doi:10.1016/j.spinee.2019.05.283

Meeting abstract from SpecialtyCare's clinical research group that classifies MEP alerts across 40,919 cervical spine procedures by neuroanatomical pattern and relates each category to the risk of a new neurologic deficit.

Publisher

Other collaborations 1

Work with other partner hospitals, including Boston Children's.

2026 · ORIGINAL RESEARCH

The Role of Intraoperative Neurophysiologic Monitoring on Surgical Decision Making and Neurologic Outcomes in Pediatric Patients With Intramedullary Spinal Cord Tumors

Papadakis JE, Mosher AM, Slingerland AL, Albanese JS, Staffa SJ, **Bose M**, **Toczylowski M**, Fehnel K
J Clin Neurophysiol. 2026;43(1):49-60. doi:10.1097/wnp.0000000000001153 · PMID 40001277

Retrospective review of 59 pediatric intramedullary spinal cord tumor resections at Boston Children's (2000 to 2022) looking at how monitoring shaped surgical decisions and outcomes. Motor evoked potential alerts often led surgeons to stop resecting. New motor and sensory deficits went with worse early outcomes, but most children had recovered function by one year. Monitoring was associated with longer stays and more discharges to rehab, without a difference in long-term outcome.

MONITORING TAKEAWAY MEP alerts frequently changed the operative plan by ending the resection.

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